COOLING ENVIRONMENTS



PG25 at Scale: What We've Learned and Where We're Going

Paul Artman - *AMD*

Chris Campbell - *Vertiv*

Jacob Paugh - *ChemTreat*

Cam Turner - *Microsoft*

Keegan Yaroch - *Dow Chemical*

General Business

Panel Discussion

Time: 8:25 – 8:50AM



Paul Artman

AMD
Fellow System Design



Chris Campbell

Vertiv
Global Product Manager
Liquid Cooling Fluid Distribution



Jacob Paugh

ChemTreat
Senior Director, Global High Tech



Cam Turner

Microsoft
Principal Technical Program Manager
Microsoft Cloud Operations and Innovation - Systems Technology Team



Keegan Yaroch

Dow Chemical
Associate TS&D Scientist
Glycol-Based Heat Transfer Fluids

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Introductions



Paul Artman is **a Fellow in System Design Engineering** at AMD. Paul has 25+ years of industry experience with a focus on enabling and scaling hardware from the node to the data center. Paul is voting member of ASHRAE TC 9.9 and is a member of the OCP AI Leadership Team.



Chris Campbell is a **Global Product Manager for Liquid Cooling Fluid Distribution Systems** at Vertiv. He has over 17 years of experience in mission-critical cooling systems; including test engineering, system architecture, and product management. Chris participates in various OCP working groups including Cold Plate, Coolant Fluids and Modular TCS. He has been a member of ASHRAE for 11 years and serves as the current secretary of ASHRAE TC 9.9.



Jacob Paugh is **Senior Director of Global High Tech** at ChemTreat, leading strategy for mission-critical markets including data centers, semiconductors, and microelectronics. He has more than 20 years of experience in industrial water treatment and specializes in thermal management, process water systems, water reuse and specialty fluid chemistries.



Cam Turner is a **Principal Technical Program Manager** at Microsoft; focused on advanced cooling, power, network and sustainability technologies for datacenters. Cam is a professional engineer with over 10 years of industry experience and has been an active contributor in OCP since 2018.



Keegan Yaroch is a Technical Service & Development Scientist at Dow with 10 years of industry experience. She specializes in glycol-based coolants for liquid-cooled data center applications and serves as co-lead of the OCP Cold Plate Coolants workstream, supporting the development of guidelines and specifications for PG25 fluids.

General Business

Updates and Call to Action

- **Base Specification for Single-Phase Propylene Glycol-Based Cold Plate Coolants (PG 25)** released in August
- **Guidelines for Using Propylene Glycol-Based Heat Transfer Fluids in Liquid-Cooled Racks** updated/released in September
- **Base Specification for Single Phase Water-Based Cold Plate Coolants** released in August
- **Cold Plate Cooling Fluids Workstream meets monthly on the second Monday.**
  - Proposals for new workstream discussions are underway

PG25 Base Specification

PG25 Guidelines

Water Base Specification

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